

Postpartum Hemorrhage 01: Why PPH Is Dangerous

Companion reading handout for simultaneous listening.

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This episode is based mainly on the Israeli position paper on early postpartum hemorrhage, the local Hadassah postpartum hemorrhage protocol and checklist, and Gabbe chapter 18, Antepartum and Postpartum Hemorrhage.

Let us begin at the bedside

The baby has delivered. The room relaxes for a moment. The drape is wet, but not dramatically wet. The uterus feels a little soft. The blood pressure is still normal. Someone says, let us watch for a few more minutes.

That sentence is where postpartum hemorrhage morbidity begins.

Not because watching is always wrong. Sometimes bleeding settles. Sometimes the uterus firms up. Sometimes the amount looks worse than it is. The problem is that postpartum hemorrhage, or PPH, is dangerous precisely because early appearance can be misleading. The patient can look stable while she is losing a clinically important amount of blood. The measured hemoglobin can still look acceptable. The blood pressure can still be normal. And yet the event may already be moving faster than the team.

So this first lecture is not about every treatment step. We will get to the medications, the Bakri balloon, embolization, surgery, hysterectomy, and massive transfusion later. This lecture is about the mental model that makes those treatments work.

PPH is a race between blood loss and source control.

Your job, as the resident in the room, is to recognize the race early, organize the response, and avoid false reassurance.

The stakes are high because severe bleeding does not injure only through low hemoglobin. The position paper and local protocol both emphasize the same pattern of maternal harm: hypovolemic shock, renal injury, respiratory failure such as acute respiratory distress syndrome, coagulation failure such as DIC, loss of fertility, and pituitary necrosis, also known as Sheehan syndrome. In other words, PPH is not simply a volume problem. It is a whole-body perfusion and coagulation crisis when it is allowed to progress.

Let us define the problem.

Early postpartum hemorrhage is bleeding that occurs within the first twenty-four hours after vaginal birth or cesarean delivery. The national position paper describes it as an obstetric emergency that can occur up to twenty-four hours after birth. The local protocol uses the older but still useful volume thresholds: more than five hundred milliliters after a vaginal delivery, or more than one thousand milliliters after a cesarean section. Severe PPH is bleeding of more than one thousand milliliters in the first twenty-four hours.

Gabbe gives an important reason not to treat those numbers as perfect walls. Average blood loss depends on the delivery. Vaginal delivery is around five hundred milliliters. Cesarean delivery is around one thousand milliliters. Cesarean hysterectomy is around fifteen hundred milliliters. So the classic thresholds are close to the expected blood loss for that mode of delivery. That is why the clinical picture matters so much. A number becomes dangerous when it is ongoing, underestimated, symptomatic, or occurring in a patient with less reserve.

But notice something important. The most clinically useful definition is not purely numeric.

PPH is also excessive bleeding that causes symptoms or signs of hypovolemia. Dizziness. Fainting. Tachycardia. Hypotension. Oliguria. Altered mental status. A patient who is bleeding and becoming symptomatic has PPH, even if the measured number is imperfect.

Gabbe makes this point very practically. A useful definition is excessive delivery-related blood loss that makes the patient hemodynamically symptomatic or hypovolemic. That is the definition that matters at the bedside, because the number in the suction canister is never the whole story.

Now, why is the number unreliable?

First, visual estimation of blood loss is notoriously poor. It usually underestimates the true amount. Blood is on pads, under the patient, in drapes, on the floor, in clots, in the bed, in the surgical field, or hidden in a hematoma. When the team says, it is probably five hundred, the actual loss may be much higher.

Second, pregnancy prepares the body to lose blood.

The local PPH protocol describes the major physiologic compensation. Plasma volume expands by about forty percent, and red blood cell mass increases by about twenty-five percent. Gabbe gives a similar range: plasma volume rises roughly forty to fifty percent, and red cell mass rises roughly twenty to thirty percent by late pregnancy.

That reserve is useful. It helps most patients tolerate normal delivery blood loss.

But it also creates a trap.

Because the patient has a larger circulating volume, she may not become hypotensive early. She can lose a meaningful amount of blood before the vital signs look dramatic.

Tachycardia may appear before hypotension. Hypotension is late. Oliguria may be late. A pale patient who says she feels faint may be telling you the diagnosis before the monitor does.

This is why PPH is not managed by waiting for the blood pressure to fall. If you wait for hypotension, you have waited for compensation to fail.

There is a second laboratory trap.

Hemoglobin is a delayed marker in acute hemorrhage.

In the first minutes of bleeding, the patient is losing whole blood. Red cells and plasma are leaving together. The concentration of hemoglobin may not immediately reflect the true loss. Later, after fluid shifts and resuscitation, the number changes. So a reassuring early hemoglobin is not permission to ignore active bleeding.

This is the first major memory point of the course.

In PPH, vital signs can lag, and hemoglobin can lag. The bleeding does not lag.

Now let us understand why uterine atony is so central

At term, blood flow through the placental site averages about five hundred to seven hundred milliliters per minute. Think about that number. Five hundred to seven hundred milliliters per minute. That means the placental bed is not a small wound. It is a high-flow vascular surface.

After the placenta separates, the uterus has to close that surface. It does this mechanically. The myometrial fibers contract and compress the spiral arterioles. Gabbe describes this as the uterus contracting in a tourniquet fashion around the vessels.

I want you to imagine the uterus as a hand closing around several open hoses. When the hand is closed, the hoses are compressed. When the hand opens, flow resumes. That is atony.

The uterus is the tourniquet.

If the uterus does not contract, bleeding can become rapid because the placental bed is still connected to a high-flow maternal circulation. This is why fundal tone is not a casual postpartum check. It is a direct assessment of the patient's primary hemostatic mechanism.

The local protocol says early PPH occurs in about four to six percent of pregnancies and is caused by uterine atony in about seventy percent of cases. Gabbe states that uterine atony is responsible for about eighty percent of postpartum hemorrhage cases. The exact percentage varies by source, but the clinical message is the same: atony is common, and every resident must recognize it immediately.

But here is the dangerous counterpoint.

The fact that atony is common does not mean every PPH is atony.

This is where the 4 T's matter.

The 4 T's are Tone, Trauma, Tissue, and Thrombin.

Tone means the uterus is not contracting. This includes atony, and sometimes inversion or retained tissue that prevents effective contraction.

Trauma means bleeding from injury. Cervical laceration. Vaginal tear. Perineal tear. Uterine incision extension at cesarean. Uterine rupture. A vulvar, vaginal, broad ligament, or retroperitoneal hematoma.

Tissue means retained placenta or membranes, or placenta accreta spectrum, where the placenta does not separate normally.

Thrombin means a coagulation problem. Congenital or acquired coagulopathy. Disseminated intravascular coagulation, or DIC. Coagulopathy from abruption, severe preeclampsia or HELLP syndrome, amniotic fluid embolism, sepsis, fetal demise, or massive hemorrhage itself.

The 4 T's are often taught as a mnemonic. That is fine, but it is not enough.

The 4 T's are not just names. They are four different treatment pathways.

If the uterus is soft and bleeding is brisk, tone is likely. Massage, empty the bladder, oxytocin, additional uterotonics, tranexamic acid, and escalation if needed.

If the uterus is firm and bleeding continues, more uterotonics will not repair a cervical tear. More uterotonics will not close a broad ligament hematoma. More uterotonics will not remove retained placental tissue. More uterotonics will not replace fibrinogen.

So the first physical question is simple.

Is the uterus firm?

If the uterus is soft, treat atony aggressively.

If the uterus is firm and bleeding continues, keep looking.

That phrase should stay with you: firm uterus, keep looking.

Now let us talk about risk factors

Risk factors matter because they help you prepare. They do not let you relax when they are absent.

The position paper lists prior PPH, high parity, previous uterine surgery, preeclampsia, an overdistended uterus, placental abruption, placenta previa, placenta

accreta, fibroids, anemia, induction or augmentation, labor dystocia, prolonged third stage, operative vaginal delivery, chorioamnionitis, precipitous labor, and prolonged tocolytic use.

That can sound like a long dry list. Let us make it useful by grouping it by mechanism.

First, there are conditions where the uterus is stretched. Multiple gestation. Polyhydramnios. Macrosomia. A stretched uterus may not contract efficiently after delivery.

Second, there are conditions where the uterus is exhausted or inflamed. Prolonged labor. Rapid labor. Chorioamnionitis. Prolonged oxytocin exposure. Induction and augmentation. These are settings where the myometrium may not respond normally.

Third, there are conditions where the placental bed is abnormal. Placenta previa. Placenta accreta. Abruption. Previous uterine surgery. These increase the risk of bleeding because the placental interface or uterine wall is not normal.

Fourth, there are conditions that make trauma more likely. Operative vaginal delivery. Shoulder dystocia. Macrosomia. Precipitous birth. These raise concern for lacerations and hematomas.

Fifth, there are conditions where the patient has less reserve or less ability to clot. Anemia. Preeclampsia. Abruption. Sepsis. Coagulation disorders.

That grouping is easier to remember because it follows the physiology. Stretched uterus. Tired or infected uterus. Abnormal placenta. Traumatic delivery. Poor reserve or poor clotting.

But the national position paper adds a crucial caution: these risk factors are common, and most women with risk factors will not have excessive bleeding.

The inverse is also true.

Some women without obvious risk factors will bleed.

So risk factors should change how ready you are. They should not determine whether you are awake.

Now let us connect this to prevention

The first preventive principle is active management of the third stage of labor, especially administration of a uterotonic after birth. The position paper notes that professional organizations recommend uterotonics, mostly oxytocin, after every birth to prevent hemorrhage. The local protocol says that in cases of increased PPH risk, active management of the third stage has a role, and that administration of ten units of Pitocin IV or I M is the most effective prevention with very few side effects.

Gabbe adds the physiologic and evidence-based frame. Active management of the third stage, spontaneous placental separation during cesarean, and prolonged postpartum oxytocin infusion all reduce atony-related bleeding. The concept is simple: prevent the living tourniquet from failing.

But prevention does not end the need for vigilance. A patient can receive oxytocin and still develop atony. She can have a firm uterus and still bleed from trauma. She can have a normal third stage and still have delayed PPH later.

So prevention lowers risk. It does not remove responsibility.

Let us now build the resident's first recognition algorithm.

When postpartum bleeding is ongoing, ask three questions.

First: is she compensating, or is compensation failing?

Look beyond the blood pressure. Is the pulse rising? Is she pale? Is she dizzy? Is she faint? Is she anxious in a way that feels physiologic? Is the urine output falling? Is she short of breath? Does she look worse than the monitor suggests?

Second: is the uterus firm?

Put a hand on the fundus. This is not a symbolic exam. You are checking whether the primary postpartum hemostatic mechanism is working. A soft uterus tells you to treat tone immediately. A firm uterus tells you not to stop thinking.

Third: have I called help?

PPH is not a solo performance. The position paper emphasizes activation of a team: midwives, nurses, anesthesiologists, blood bank and laboratory personnel. The local protocol emphasizes rapid communication with anesthesia, nursing, and the obstetric team. Early help is not a sign that you are losing control. It is how control is maintained.

This is the second major memory point.

One hand on the uterus. Two large lines. Three calls. Four T's.

The exact sequence will vary by room and severity, and we will go through it in detail in the next episode. But the structure is already there.

One hand on the uterus means assess and treat tone.

Two large lines means create access before the patient crashes.

Three calls means bring senior obstetric help, anesthesia, and blood bank into the case early.

Four T's means diagnose the mechanism while treatment begins.

Now let us clarify early and delayed PPH, because this distinction prevents confusion

Primary, or early, PPH occurs within twenty-four hours. This is the classic delivery-room emergency: atony, laceration, retained placenta, accreta, rupture, inversion, or coagulopathy.

Secondary, or delayed, PPH occurs after twenty-four hours and up to twelve weeks postpartum. The local protocol lists subinvolution of the placental bed, retained placenta, uterine infection, and congenital coagulation disorders, such as von Willebrand disease. Gabbe notes that delayed bleeding requiring medical attention occurs in about one to two percent of patients, and often occurs between days eight and fourteen of the puerperium.

We will devote a later episode to delayed PPH. For now, remember the boundary: the first twenty-four hours is early PPH. After that, think delayed PPH, with a different differential and a different tempo.

Let us return to the delivery room and make this concrete.

A patient delivers vaginally after a long induction. The placenta delivers. Ten minutes later, there is a steady gush. The uterus is boggy. The measured loss is four hundred milliliters, but the bleeding is still active.

Do not wait until the number becomes five hundred to begin thinking. The mechanism is already visible. A stretched, tired, or oxytocin-exposed uterus may be failing to contract. The correct thought is not, the blood pressure is normal, so this is fine. The correct thought is, this is active bleeding with probable atony, and the placental bed can bleed fast.

Now change the case.

A patient delivers by vacuum. The uterus is firm, but bleeding continues. The perineum looks repaired, but the pad keeps filling. This is where residents get trapped if they keep repeating uterotonics because PPH equals atony in their mind. A firm uterus with ongoing bleeding should make you look for cervical laceration, vaginal sidewall injury, hematoma, or another non-tone source.

Change the case again.

A patient has severe preeclampsia and abruption. The uterus may be tender. The bleeding may be partly concealed. The blood pressure may be high at first, not low. Then the labs return with falling fibrinogen and platelets. This is a thrombin problem as much as a bleeding problem. More uterine massage alone will not fix DIC.

These cases teach the same point from different angles.

Do not memorize PPH as a single disease. PPH is a clinical syndrome with several mechanisms.

Now, what are the specific report thresholds in the local protocol?

Bleeding over five hundred milliliters in a parturient who continues to bleed should be reported.

Any bleeding over one thousand milliliters should be reported.

Bleeding over five hundred milliliters in an anemic patient, with hemoglobin below nine grams per deciliter before delivery, should be reported.

And bleeding over five hundred milliliters with symptoms of hypovolemic shock, such as tachycardia, a drop in blood pressure, or altered consciousness, should be reported.

Do not hear those thresholds as bureaucracy. Hear them as safety rails. They exist because the human eye underestimates blood loss, and because a resident under pressure can normalize a situation that is slowly becoming dangerous.

Let us now summarize the key physiology.

Pregnancy gives the patient reserve. That reserve is protective, but it delays obvious deterioration.

The uterus is the main postpartum hemostatic organ. Its contraction compresses the placental bed.

Atony is common, but not universal.

The 4 T's convert bleeding into a treatment map.

And early recognition depends on action before collapse.

I want to end this first episode with the exact cognitive reflex I want you to carry into the rest of the course.

When postpartum bleeding is ongoing, do not ask only, how much blood is this?

Ask, is she compensating?

Ask, is the uterus firm?

Ask, what T is this?

Ask, have I called help?

Because PPH does not punish the resident who starts organized treatment early. It punishes the team that waits for the situation to become obvious.

In the next episode, we will take this recognition framework and turn it into the first five minutes of action: who to call, what to say, what to do with your hands, what to send to the lab, when to involve blood bank, and how to keep resuscitation and diagnosis moving at the same time.

For now, lock in four takeaways.

First, postpartum hemorrhage is a race between blood loss and source control.

Second, normal vital signs and early hemoglobin can falsely reassure you.

Third, the uterus is the living tourniquet, but a firm uterus means you must keep looking.

And fourth, the 4 T's are not just a mnemonic. They are the map that tells you what treatment will actually stop the bleeding.